Endogenous Growth

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Based on Julia K. Thomas, Topic 4

Motivation

Kaldor facts: what should a long-run growth model explain?

Economic data over long periods (especially for the U.S. and U.K.) suggest:

1. roughly constant capital and labor shares of national income,
2. roughly constant growth rate of the capital stock,
3. roughly constant growth rate of output per worker,
4. roughly constant capital–output ratio,
5. roughly constant rate of return on investment, and
6. steadily growing real wages.

Implication. A useful growth model should deliver a balanced growth path:

- factor shares are stable,
- aggregate quantities grow at regular rates,
- the return to capital is stable, and
- wages rise with productivity.

Kaldor facts in a neoclassical growth model

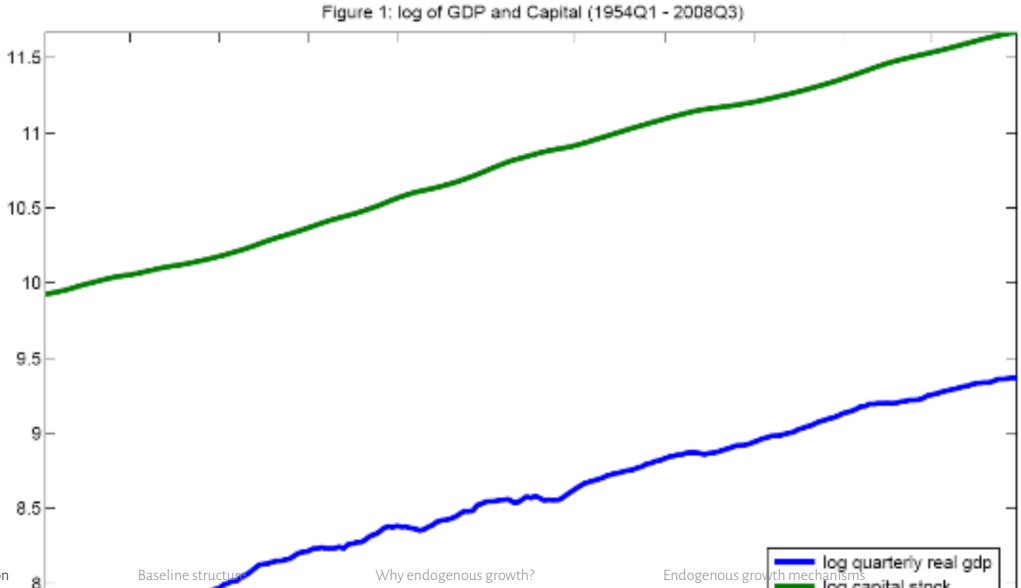
Kaldor's observations lead us to emphasize balanced growth paths with:

- constant factor shares,
- capital and output growing at a common rate,
- output per worker and capital per worker moving together,
- a constant rental rate on capital, and
- growing real wages.

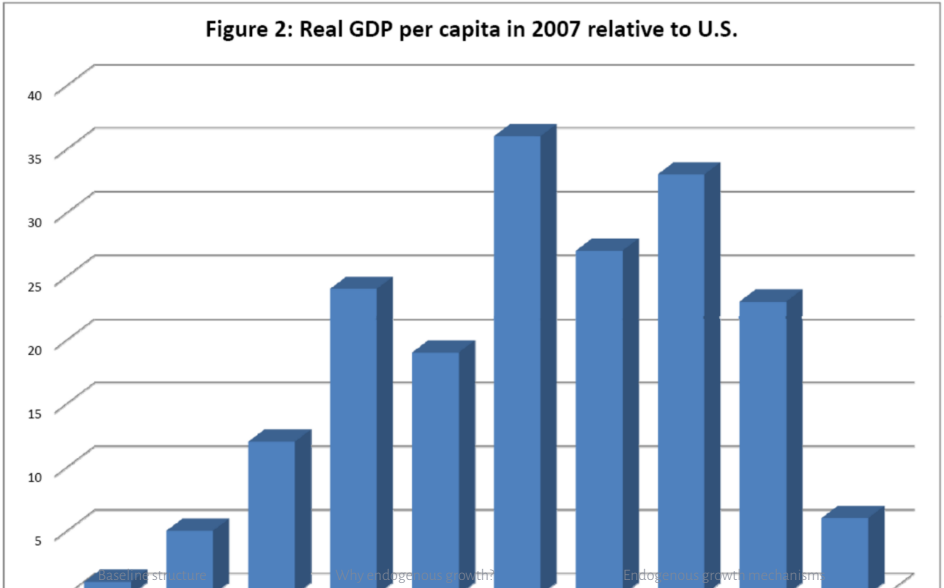
These observations guide common functional-form choices:

1. **Cobb–Douglas production**: helps deliver stable factor shares.
2. **Separable CES preferences**: helps support balanced growth with stable labor supply behavior.
3. Together with common growth in Y and K, these help generate growing real wages.

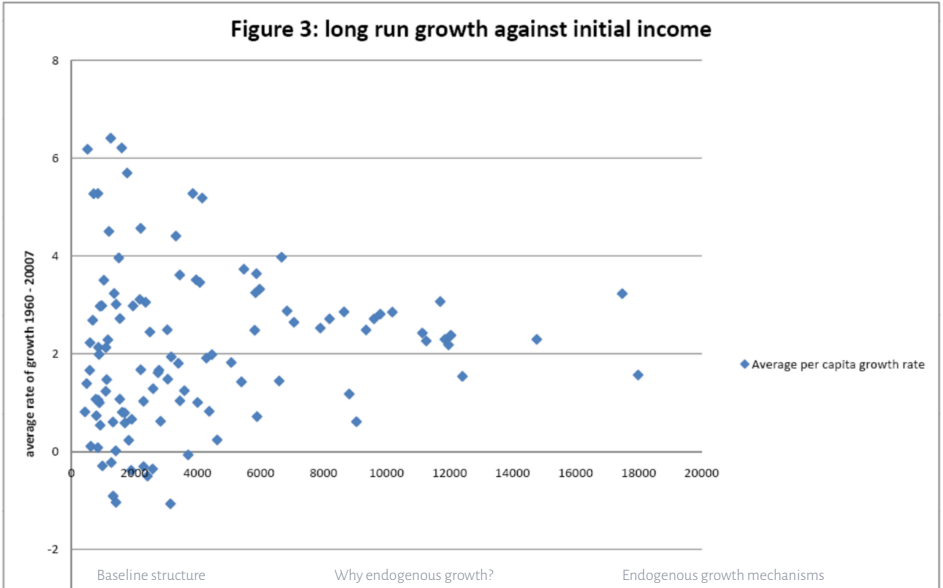
Exact figure from the source: capital-to-GDP roughly constant despite growth



Exact figure from the source: dispersion in per-capita income levels



Exact figure from the source: dispersion in per-capita growth rates



Development accounting: where do income differences come from?

Suppose production is CRS in physical capital and efficiency units of labor:

$$Y = K^{\alpha}(AhL)^{1-\alpha}.$$

Then output per person can be decomposed as

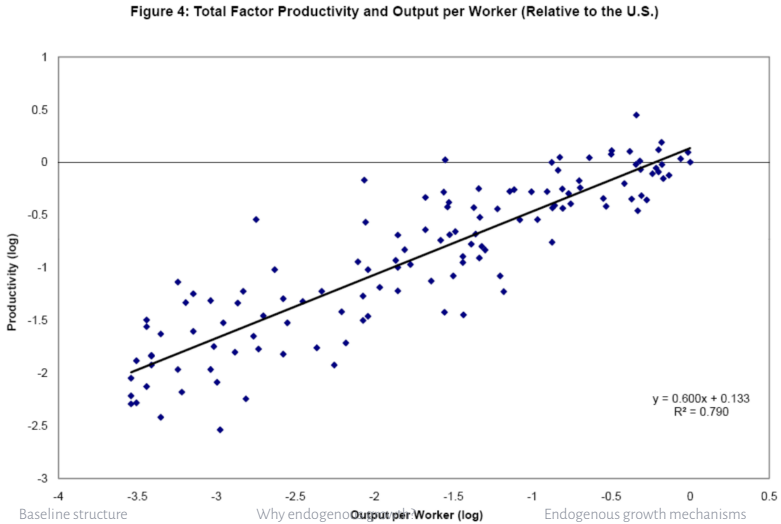
$$\frac{Y}{\text{pop}} = \left(\frac{K}{Y}\right)^{\frac{\alpha}{1-\alpha}} \left(\frac{L}{\text{pop}}\right) hA.$$

- $\frac{K}{Y}$ captures capital intensity.
- $\frac{L}{\text{pop}}$ captures labor-force participation / hours.
- h captures human capital.
- A captures total factor productivity (TFP).

A key empirical lesson from development accounting is that TFP differences explain a large share of cross-country income gaps.

Exact figure from the source: TFP's role in cross-country income differences

from Hall and Jones (1999)



Baseline structure

Structure housing endogenous growth models

Households

- A constant large number of identical households (often normalized to $N = 1$).
- Isoelastic utility over consumption:

$$u(C) = \frac{C^{1-\sigma} - 1}{1 - \sigma}, \qquad \sigma > 0.$$

- No valuation of leisure in the baseline version.
- Lifetime objective:

$$\max \sum_{t=0}^{\infty} \beta^t u(C_t).$$

- Capital accumulation:

$$K_{t+1} = (1 - \delta)K_t + I_t.$$

Structure housing endogenous growth models (continued)

Final-goods producers

- Perfect competition in input and output markets.
- CRS production:
$$Y = F(K, X),$$
where X is overall labor input or efficiency units of labor.
- Under CRS, we can write
$$F(K, X) = Xf(k), \qquad k \equiv \frac{K}{X}.$$
- Competitive factor prices are:
$$r + \delta = f'(k), \qquad w = f(k) - kf'(k).$$

Implication. With perfect competition and CRS, total output is exhausted by factor payments, so profits are zero in equilibrium.

Why endogenous growth?

Case 1: fixed labor-augmenting skills

Suppose

$$Y_t = F(K_t, AN),$$

where A is constant.

- If A is fixed, then as K_t rises, the capital–efficiency-labor ratio rises.
- Because $f'(k)$ is decreasing, the marginal product of capital falls.
- The Euler equation is
$$u'(C_t) = \beta u'(C_{t+1})\left[f'(k_{t+1}) + 1 - \delta\right].$$
- Sustained per-capita growth would require returns to remain high enough forever.
- But diminishing returns imply the economy converges to a steady state with no long-run per-capita growth.

So the standard neoclassical model needs exogenous technological progress to explain persistent growth.

Case 2: exogenously growing skills

Now let

$$X_t = A_t N, \qquad A_t = \gamma_A A_{t-1}, \quad \gamma_A > 1.$$

Then

$$Y_t = F(K_t, A_t N) = A_t N f\left(\frac{K_t}{A_t N}\right).$$

- If K_t and A_t grow at the same rate, then $k_t = K_t/(A_t N)$ stays constant.
- The marginal product of capital stays constant as well.
- Under isoelastic utility, the Euler equation implies

$$\gamma_C^\sigma = \beta\left[f'(k) + 1 - \delta\right].$$

This delivers balanced growth, but growth is still exogenous because A_t grows outside the model.

Endogenous growth mechanisms

The AK model: the simplest endogenous growth model

Rebelo's AK model assumes

$$Y = AK.$$

- The marginal product of capital is constant: $MPK = A$.
- There are no diminishing returns to the accumulable factor.
- The Euler equation becomes

$$\gamma_C^{\sigma} = \beta\left[A + 1 - \delta\right].$$

- If $\beta[A + 1 - \delta] > 1$, then consumption, capital, and output all grow forever.

Why is this important?

- It satisfies many Kaldor facts: constant return to capital, constant K/Y, and constant output growth.
- Policy can affect long-run growth directly.

Policy matters in the AK model

Suppose output is taxed at rate τ, so disposable output is

$$Y^D = (1-\tau)AK.$$

Then the Euler equation becomes

$$\gamma_C^\sigma = \beta\left[(1-\tau)A + 1 - \delta\right].$$

- A higher output tax lowers the private return to accumulation.
- Therefore it lowers the economy's long-run growth rate.
- This is a central distinction between exogenous-growth and endogenous-growth models:
  - in exogenous-growth models, policy often affects levels but not long-run growth;
  - in endogenous-growth models, policy can affect the growth rate itself.

Learning by doing / external effects

Romer (1986) generates endogenous growth through a positive externality.

- Let production be $Y_t = F(K_t, X_t)$, with $X_t = A_t N$.
- In equilibrium, knowledge evolves with aggregate capital, e.g.

$$A_t = \frac{K_t}{N},$$

  but each firm takes A_t as given.
- Privately, the firm sees CRS and chooses capital competitively.
- Socially, capital accumulation also raises knowledge, so there is an external return to investment.

Implications

1. constant growth in consumption and GDP is possible,
2. policy can affect growth,
3. the competitive equilibrium is generally not efficient because firms ignore the knowledge spillover.

Takeaways

1. Kaldor facts motivate balanced-growth analysis.
2. Development accounting suggests that TFP is central for understanding income differences.
3. With fixed technology, the neoclassical model does not generate sustained per-capita growth.
4. Exogenous growth can be introduced through growing A_t, but that leaves the source of growth unexplained.
5. Endogenous growth models explain long-run growth from economic forces inside the model:
   - AK: constant returns to accumulable capital,
   - learning by doing: external effects of accumulation,
   - later models: R&D, product variety, quality ladders, and human-capital accumulation.